How to Read, Write, and Summarize a Computer Science Research Paper

An Introduction for 2nd Year Students

Research Methodology Module

Department of Computer Science

Spring Semester 2026

Roadmap for Today

The Mindset: Research is a Conversation

Methodology: The Three-Pass Method

Anatomy of a CS Research Paper

Live Dissection: Anatomy of a Seminal Paper

Writing the Summary: The Final Synthesis

The Mindset: Research is a Conversation

Why Are Research Papers Important?

- **They are the edge of knowledge.** Textbooks are 5-10 years behind.
- **They are formal conversations.** Researchers proposing ideas, disputing claims, and validating results.
- **As a CS Sophomore:** You have the programming/math foundations to understand *what* problem is being solved, even if the math/architecture is complex.

The Core Questions to Ask (Every Time)

1. What bottleneck/problem are they solving?
2. Why is this problem *hard*?
3. What is their key, clever insight?

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Methodology: The Three-Pass Method

How (Not) To Read a Paper

Warning

Do *not* read a research paper linearly from page 1 to the end. It is inefficient and frustrating.

Instead, adopt a structured approach. The goal is increasing depth with each pass.

The Three-Pass Reading Method: Illustrated

This custom diagram visualizes how each pass increases comprehension and depth.

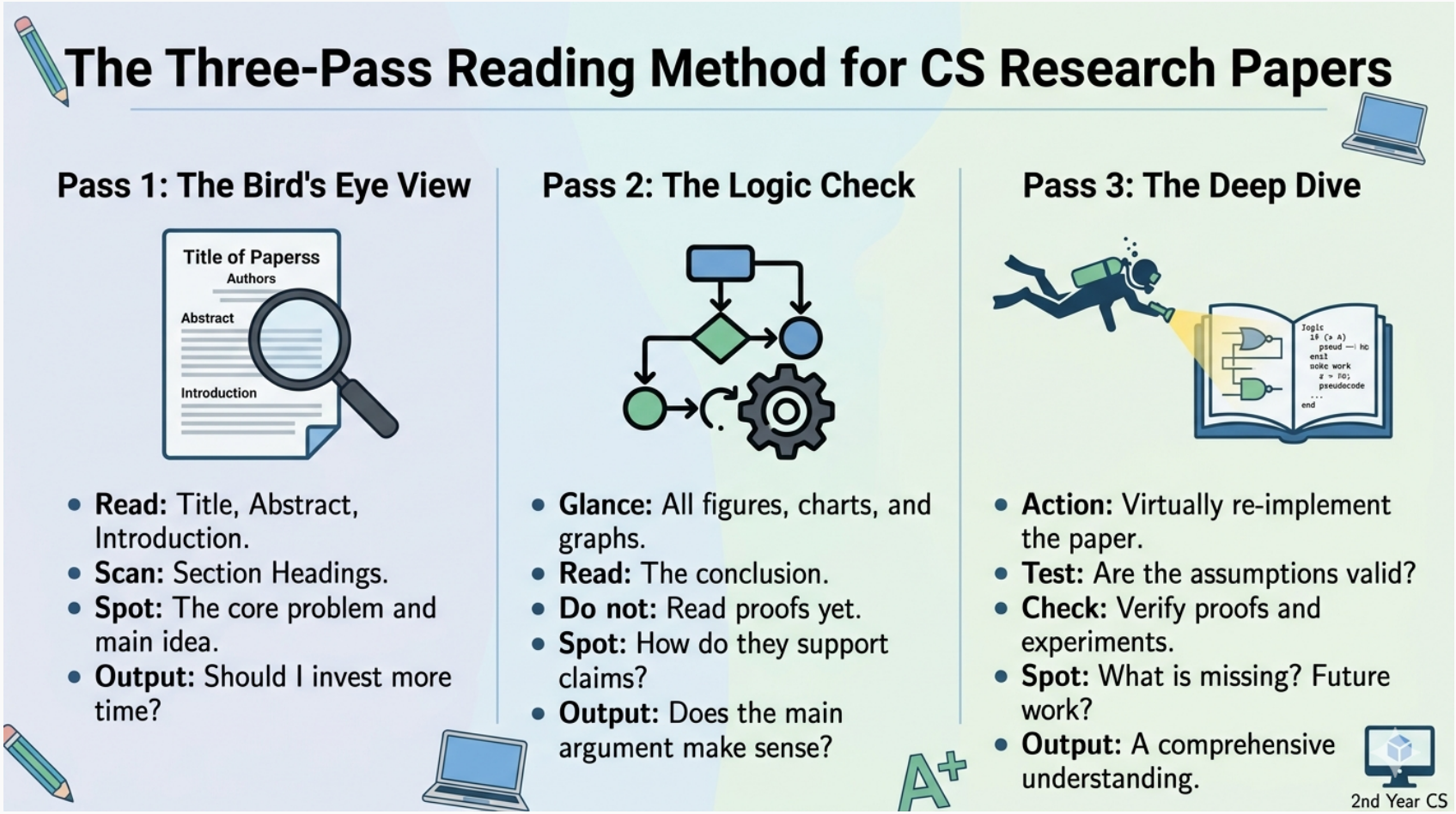


Figure 1: A roadmap for strategic reading. © 2nd Year CS Module.

Deconstructing the Passes

Pass 1: The Bird's Eye View (5-10 mins)

- Read: Title, Abstract, Introduction.
- Scan: Headings.
- **Output:** Decide if the paper is relevant enough to keep reading.

Pass 2: The Logic Check (1 hour)

- Examine: All Figures and Graphs.
- Read: Conclusion.
- **Output:** Understand *how* the argument works. (Skip proofs/code for now.)

Pass 3: The Deep Dive (2+ hours)

Virtually re-implement the paper. Check every assumption. Re-derive the proofs.

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Anatomy of a CS Research Paper

Anatomy of a Paper (Standard CS Structure)

Title/Abstract The 'hook.' Summarizes problem, insight, and result.

Introduction The 'sales pitch.' Context, motivation, and main contributions.

Related Work Where this paper fits in the conversation. Who else tried this?

System/Method The core contribution. Architecture diagrams, algorithms, or mathematical model.

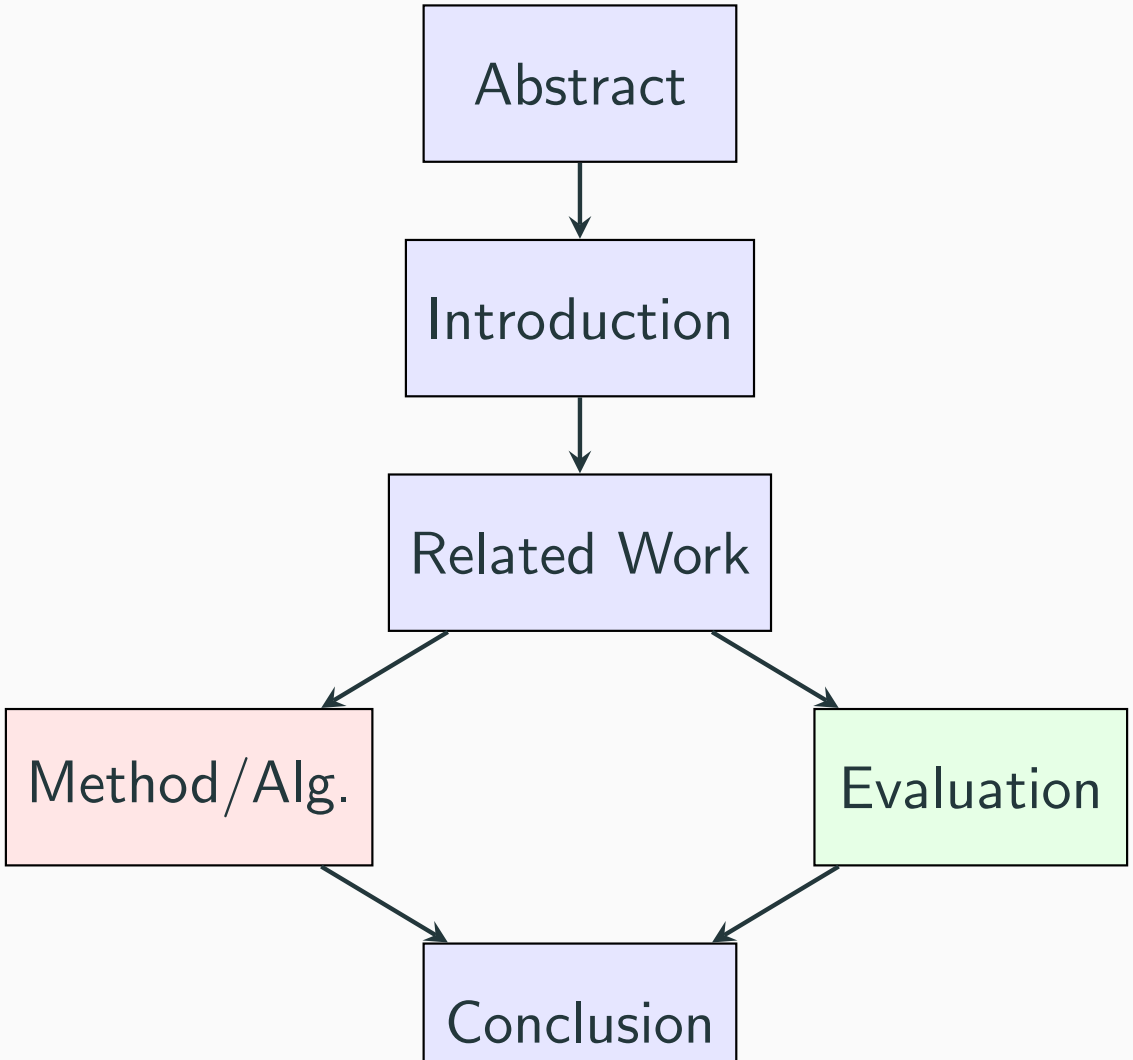
Evaluation The proof. Experiments, benchmarks, complexity analysis ($O(n)$).

Conclusion Future work and summary of impact.

Live Dissection: Anatomy of a Seminal Paper

The Dissection Protocol

We will map the 'Three-Pass Method' onto a concrete example using this simplified structure flow:



The Live Example: Google's PageRank (1998)

The Paper

"The PageRank Citation Ranking: Bringing Order to the Web"
by Lawrence Page and Sergey Brin (Stanford, 1998).

Why this paper?

- It founded Google.
- The solution is elegant and uses basic Discrete Math (Graphs).
- Perfect for 2nd-year CS students.

Pass 1: Dissecting the PageRank Abstract

- **Problem (The Hook):** The exploding size of the web makes traditional keyword search useless. Relevant results get lost.
- **Insight:** Treat the web as a directed graph. A link from page A to page B is a 'vote' of confidence for B.
- **Method:** PageRank simulates a "random surfer" who clicks links. A page's importance is proportional to the probability that a random surfer lands on it.

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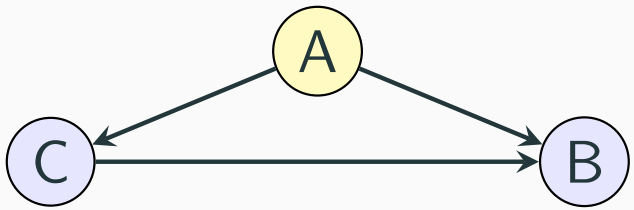
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Pass 2: Visualizing the Insight (Figure 1)

We scan the paper for the key visual aid.



Low PR High PR

Dissecting Figure 1:

- A simple directed graph.
- Yellow Node A has links from both B and C (many incoming votes).
- Node A is 'important.'
- Its votes *back* to B and C carry more weight.

Pass 3: Deep Dive into the Algorithm

Now, and only now, do we tackle the math we saw in the "Method" section.
The heart of PageRank is a recursive definition:

Simplified Formula (from the paper)

$$PR(A) = (1 - d) + d \left(\frac{PR(T1)}{C(T1)} + \cdots + \frac{PR(Tn)}{C(Tn)} \right)$$

$PR(A)$ The PageRank of page A.

$C(Ti)$ The number of outgoing links on page Ti.

d The 'damping factor' (usually 0.85). The chance the surfer keeps clicking.

Comparing Anatomy to Execution (Review)

- **Anatomy (What we expected):** Intro $\rightarrow$ Method $\rightarrow$ Evaluation.
- **Execution (What we did):**
  1. Identified problem in Abstract.
  2. Found the intuition in Pass 2 Figures.
  3. Verified logic in Pass 3 via the recursive equation.

= SECTION 5: SUMMARIZATION

Writing the Summary: The Final Synthesis

Summarization Template (Your Next Assignment)

Do not just summarize the paper's abstract. Summarize *your understanding* of the paper. Use this four-step synthesis structure:

1. **Problem:** What specific bottleneck did they attack?
2. **Key Insight/Method:** What clever intuition solved it? (Draw the simplified diagram).
3. **Validation:** What metrics (Throughput, Latency, Accuracy) did they use? What were the key results?
4. **Critique/Context:** What did they miss? How does this connect to your own coursework?

Key Takeaways

- Reading is a **strategic attack**, not a slow crawl.
- Follow the **Three-Pass Method** to save time.
- **Figure 1** is often the entire paper summarized visually. Tackle it in Pass 2.
- **Deep Dive math** is saved for the end (Pass 3).
- You *already have the tools* to understand seminal ideas.

Questions?